Student Project Management System (SPMS)

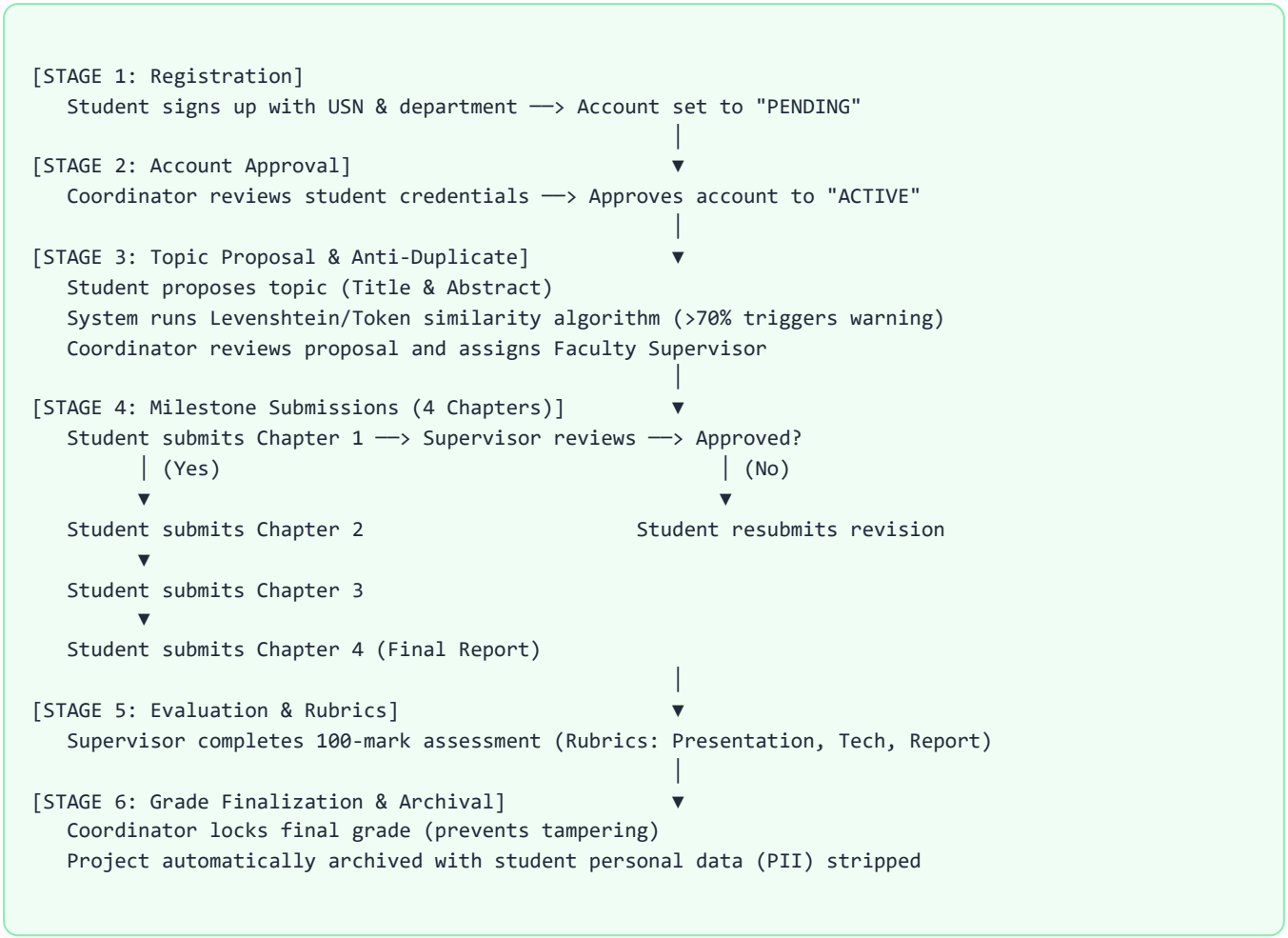
Comprehensive Guide to System Flow, Architecture, and Role Spaces

Version 1.0.1 • Fast, Monolithic Python/FastAPI Application with MySQL 8.0

> **Executive Summary:** SPMS is a monolithic web platform designed to streamline, govern, and evaluate final-year engineering capstone projects. It enforces role separation among **Coordinators (Administrators)**, **Faculty Supervisors**, and **Students**, covering the complete lifecycle from topic proposal and plagiarism screening to milestone reviews, rubric scoring, and final PII-stripped archiving.

1. End-to-End Project Lifecycle Flow

The progression of a capstone project moves sequentially through six structured stages:



2. The 4-Chapter Milestone Pipeline

Milestone	Focus Area	Student Deliverable	Supervisor Review Criteria
Chapter 1	Introduction & Problem Formulation	PDF/DOCX with background, motivation, scope, and objectives.	Feasibility, problem clarity, realistic technical scope.
Chapter 2	Literature Review	Survey of existing research, related IEEE/ACM papers, comparative study.	Depth of citations, critical analysis of existing solutions.
Chapter 3	System Design & Methodology	Architecture diagrams, DFDs, ERDs, algorithm formulations, tech stack.	Architectural soundness, database design correctness.
Chapter 4	Implementation & Final Report	Complete software artifact, source code ZIP, final spiral report.	Working demonstration, test results, documentation completeness.

3. Workspaces & Responsibilities by Role

1. Student Space

ROLE: STUDENT

Students use their workspace to track their project progress, submit work against deadlines, and interact with their assigned guide.

Page / Feature	URL Route	Capabilities & Purpose
Student Dashboard	`/dashboard`	Visual 4-stage progress tracker, active supervisor contact card, upcoming deadline warnings, and unread notifications.
My Profile	`/profile/student`	View registered USN, department, semester, academic session, and securely change password.
Topic Proposal	`/topics`	Submit a new capstone topic proposal (Title, Abstract, Scope) or browse and select faculty-published topics. Automated duplicate screening flags similar titles in real time.
Submissions Pipeline	`/submissions`	Access the 4 submission cards (Chapters 1 to 4). Upload files (.pdf, .docx, .zip), track submission review state, read supervisor feedback notes, and upload revisions.
Supervisor Chat	`/messages`	Private, real-time 1-on-1 messaging channel with the assigned faculty supervisor for guidance and review comments.
Department Forum	`/forum`	Public department-wide discussion board to ask questions, view announcements, and exchange ideas with peers and faculty.
Guidelines	`/guidelines`	Download official university templates, standard report formatting requirements, and rubric sheets.
Project Archive	`/archive`	Search past years' approved projects by keyword, domain, or technology to research prior work and ensure topic originality.

2. Faculty Supervisor Space

ROLE: SUPERVISOR

Supervisors mentor assigned students, review chapter submissions iteratively, and conduct academic evaluations.

Page / Feature	URL Route	Capabilities & Purpose
Supervisor Dashboard	`/dashboard`	Summary metrics: total assigned students, submissions pending immediate review, unread messages, and supervisor slot usage.
Published Topics	`/topics/published`	Create and publish departmental research topics that students can view and choose for their capstone.
Review Submissions	`/submissions/reviews`	The central grading and feedback pipeline. Inspect student file uploads, review historical version revisions, write inline remarks, and click **Approve** or **Request Changes**.
Student Messages	`/messages`	Direct communication workspace with all assigned supervisees with conversation threads.
Department Forum	`/forum`	Post announcements, interact with department students, and respond to public questions.
Assessments	`/assessments`	Unlockable once Chapter 4 (Final Report) is approved. Enter evaluation scores (0–100) across presentation, methodology, and documentation rubrics.
Project Archive	`/archive`	Browse past projects and access prior research documents.

3. Coordinator / Administrator Space

ROLE: ADMIN

The Coordinator acts as the central department administrator governing user accounts, topic assignments, schedule deadlines, and final grade certification.

Page / Feature	URL Route	Capabilities & Purpose
Admin Dashboard	`/dashboard`	Department-level overview: total students, active projects, pending student registrations, and stage completion stats.
Account Approvals	`/coordinator/approvals`	Review pending student registrations. Verify institutional USN and student details, then Approve or Reject registration.
Supervisor Slots	`/coordinator/supervisors`	View faculty supervisors, monitor student load, and configure maximum supervision slot limits (e.g. 5 projects).
Topic Approvals	`/coordinator/topics`	Review student proposals, inspect algorithmic duplicate warnings, approve/reject topics, and assign an eligible supervisor.
Assessments & Grades	`/coordinator/assessments`	Review supervisor evaluations, add official coordinator remarks, and **Finalize** the project grade (permanently locks grades).
Deadlines Management	`/coordinator/deadlines`	Set and modify submission deadlines for each of the 4 project milestones. Submissions past deadline are flagged late.
Departments & Sessions	`/coordinator/departments` `/coordinator/sessions`	Configure institutional academic departments (e.g., CSE, ISE, ECE) and manage active academic calendar years/semesters.
Guidelines Upload	`/guidelines`	Upload official project handbooks, formatting templates, and evaluation criteria for students and supervisors.

4. Demo Account Credentials (Ready to Test)

Role	Username / USN	Password	Current State / Data
Coordinator / Admin	`coord_demo`	`Password123!`	Full governance, approvals pending, finalization access.
Faculty Supervisor	`prof_turing`	`Password123!`	Dr. Alan Turing (CSE Department, 5 slots). Active supervisees.
Faculty Supervisor	`prof_hopper`	`Password123!`	Dr. Grace Hopper (ISE Department, 5 slots).
Student	`1SP23CS001`	`Password123!`	Ada Lovelace (Active project in progress).
Student	`1SP23CS002`	`Password123!`	Charles Babbage (Proposed topic awaiting approval).
Student	`1SP23CS003`	`Password123!`	Margaret Hamilton (Completed and Archived project).

5. How to Run the App in VS Code

> **Terminal Command (PowerShell):**
> ```
> .\.venv\Scripts\Activate.ps1
> uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
> ```
>
> Open your browser at: **http://127.0.0.1:8000**